

Ahmed Elkhrahy

Engineer-In-Training (EIT)

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EDUCATION

B.Sc. Mechanical Engineering, Minor in Mechatronics (GPA: 3.8)

Expected April 2026

Schulich School of Engineering, University of Calgary

- **Academic Awards:** Schulich Research Award; President's Admission Scholarship; Dean's List (4 years).

TECHNICAL SKILLS

Software: SolidWorks, Aspen HYSYS, ANSYS, AutoCAD, Revit, MATLAB/Simulink, Power BI, SAP PM, AVEVA PI, Advanced Excel, Microsoft Suite

Methods & Standards: Finite Element Analysis (FEA), Risk-based Inspection (RBI), Failure Modes and Effects Analysis (FMEA), SCADA data analysis, Process Simulation, P&ID / PFD interpretation, ASME B31.3, API 580, IEC 61400

Professional: Contractor coordination, technical documentation, team leadership, project scheduling, safety-oriented

WORK EXPERIENCE

Enbridge – Power Operations | Calgary, AB

May 2024 – Sep 2025

Reliability and Integrity Co-op Engineer

- Used **AVEVA PI SCADA** to trend **power output curves**, **nacelle temperatures**, **rotor RPM**, and **vibration signatures** across a fleet of ~**300 wind turbines** (~**900–1,200 MW** installed capacity) and solar assets to detect anomalies and drive corrective maintenance decisions.
- Tracked **solar PV inverter string performance** via **irradiance-normalized yield**, **DC/AC ratios**, and **inverter fault codes** to identify underperforming strings and prioritize maintenance interventions.
- Built **Power BI** dashboards integrating **SCADA** and **SAP PM** data to monitor equipment availability and inspection KPIs; cut reporting time by ~**40%** and presented findings to senior engineers and operations leadership.
- Coordinated **wind blade inspection campaigns**; Interpreted **P&IDs**, and **PFDs** for wind and solar facilities to define inspection scope boundaries; classified **leading edge erosion**, **delamination**, and **lightning strike damage** per **IEC 61400** and initiated corrective work orders.
- Investigated **gearbox bearing degradation**, **pitch actuator faults**, and **yaw system misalignment** using SCADA fault logs, SAP histories, and **lube oil analysis** to identify failure drivers and recommend reliability improvements.
- Developed **equipment criticality matrices** for wind turbine **main bearings**, **gearboxes**, **generators**, and **solar inverters** to risk-rank assets and optimize **preventive maintenance** frequency and inspection scope across the renewable fleet.
- **Enbridge Intern of Merit Award** — Top 15% of co-op students.

Carbon Fiber Manufacturing Laboratory | Calgary, AB

May 2023 – Dec 2023

Research Intern

- Built **SolidWorks** thermal models of a **graphitization furnace** to map **heat flux distribution** across carbon fiber tow bundles, revealing **15–20% temperature non-uniformity** across the furnace cross-section that was linked to inconsistent fiber microstructure, findings used to recommend revised **zone heating setpoints and ramp rates**.
- Designed and assembled a **microwave plasma reactor** as an alternative to conventional furnace heating; benchmarked **energy efficiency**, **activation uniformity**, and **fiber tensile properties** against baseline results.

SOAR – Schulich Rocketry Engineering Club | Calgary, AB

Aug 2023 – Apr 2025

Materials & Structural Team Lead

- Led a 8-person structural subteam through full **design-build-fly** cycle; won **1st place** at **Spaceport America Cup (SAC) 2023** — world's largest intercollegiate rocketry competition with **150+ teams** across **20+ countries**.
- Evaluated **fiberglass**, **carbon fiber**, and **hybrid composite layups** against flight loads and mass budgets; ran **ANSYS FEA** on **axial thrust**, **aerodynamic bending**, and **landing impact** cases to eliminate stress concentrations before fabrication.
- Fabricated and inspected composite airframe sections using **vacuum bagging** and **wet layup**; conducted **post-cure dimensional inspections** to certify structural integrity before flight.

PROJECTS

Capstone – Compressor Station VR Trainer | Calgary, AB

Sep 2025 – Apr 2026

- Placed **3rd out of 149** competing teams at the University of Calgary Capstone Design Fair for the development of an immersive virtual reality training platform for **compressor station operations** and emergency response.
- Built **Aspen HYSYS** simulations of a natural gas compressor station — **centrifugal compressor performance curves**, **interstage cooling**, **anti-surge control logic**, and **pipeline pressure drop** — validated across variable load and turndown conditions.
- Studied system behavior under **surge onset**, **blocked discharge**, and **ESD activation** fault scenarios to define **emergency response procedures** embedded directly into operator training modules.
- Integrated a **multi-agent AI instructor** into a **VR environment** for real-time **procedural guidance** and **performance feedback** — eliminating the need for live instructors or plant shutdowns during training.
- Addressed a real industry gap: enabled **safe, repeatable training** on **high-consequence fault events** that cannot be replicated on live equipment, reducing operator training risk and cost.